Class 17: Fermi-Dirac Statistics: Part 1

As the name suggests, this distribution is attributed to Enrico Fermi, and Paul Dirac, as a result of their work in 1926. Particles that display Fermi-Dirac statistical predictions are referred to as 'Fermions'. Fermi-Dirac statistics describes the distribution of a collection of Fermions across energy levels for a system in thermal equilibrium. For our purposes Fermi-Dirac statistics is of relevance since electrons qualify as Fermions. The use of this statistical distribution will enable us to decide what the electrons can do, can't do, and will tend to do, in the solid.

Fermions follow quantum mechanical rules, are identical and indistinguishable, and specifically adhere to the Pauli's exclusion principle which states that no two particles can occupy the same quantum state at the same time. One of the requirements that a particle needs to satisfy to be a Fermion, is that it should have a half integer spin. Electrons satisfy these requirements and are therefore Fermions. The criterion of being identical and indistinguishable implies that we are unable to distinguish between cases where electrons are swapped between energy levels, leaving the total number of electrons at each energy level unchanged. We are only able to distinguish between situations where the numbers of electrons occupying a state changes.

Within the framework of electrons being fermions, and the total number of particles being constant, the total volume of the system being constant, and the total energy of the system being constant, we wish to find out the probability that a state with energy ε is occupied.

Let us consider a system of particles that are allowed to occupy energy levels

$$\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \ldots \varepsilon_i.$$

Let the number of states at these energy levels be:

$$s_1, s_2, s_3, s_4, \ldots s_i \text{ respectively.}$$

These energy levels as well as the number of states at these energy levels are fixed for the system. What values these are fixed at is a topic we will take up at a later stage. For the moment we simply accept that they are fixed.

The variable we have at our disposal is the number of particles that will occupy these states. Let the number of particles at the respective energy levels be:

$$n_1, n_2, n_3, n_4, \ldots n_i$$

A microstate of the system is then defined as a specific set of numerical values for n_1, n_2, n_3, $n_4, \ldots n_i$. A different microstate will have a different set of values for n_1, n_2, n_3, $n_4, \ldots n_i$. In the analysis that we will carry out, the values of s_i and ε_i are fixed for the system, since we have defined or identified a system with those values. The value of n_i is however

variable, and it is our intention to find out what will be the equilibrium values of n_i, at the s_i states at each of the ε_i, given the various constraints we are placing on the system. More than focusing on the exact numerical value of n_i, we will aim to determine the probability of occupancy of a state at energy level ε_i, which will be based on a ratio of n_i to s_i, which will be a more useful parameter.

In our analysis to find the probability of occupancy of a state at energy level ε_i. we will approach the calculations as follows:

1) We will start from a narrow starting point which isolates the type of term we are interested in.
2) We will then build a general equation that appears to handle the system in its entirety.
3) On examining this general equation we will note that only one term in the equation is of interest to us, so we will neglect the rest of the terms and focus only on that term from then on for the rest of the derivation.

Conventionally, the Fermi-Dirac distribution is denoted by $f(\varepsilon)$ which represents the probability that the energy level ε is occupied. Therefore, our goal is to determine an expression for $f(\varepsilon)$.

We begin by noting that each possible state of the system is characterized by a specific set of n_1, n_2, n_3, n_4,…n_i, which are the number of particles occupying energy levels ε_1, ε_2, ε_3, ε_4,...ε_i respectively. We shall denote an individual microstate of the system and its specific combination of n_1, n_2, n_3, n_4,…n_i, by $\bar{N}$ = (n_1, n_2, n_3, n_4,…).

Our narrow starting point is that when the system is in the specific individual state $\bar{N}$, the probability that a state at energy level ε_i is occupied is referred to as the 'conditional probability', since it is conditional on the system being in the individual microstate $\bar{N}$. Once an $\bar{N}$ is chosen, all of the values ε_i, s_i, and n_i are fixed for the system. Therefore, under this specific condition, the probability that a state at energy level ε_i is occupied is simply $\frac{n_i}{s_i}$

This conditional probability is denoted by $f(\varepsilon_i|\bar{N})$

So for example if at a given energy level there are 10 states, and there are 4 particles at that energy level due to a specific choice of $\bar{N}$then, under those conditions, the conditional probability of occupancy of a state at that energy level is 40%

If a different $\bar{N}$ were to be chosen which resulted in 7 particles at that energy level , then under the conditions of the new $\bar{N}$, the conditional probability of occupancy of a state at that same energy level is 70%

From this conditional probability that a state at a given energy level is occupied, how do we get the general probability that a state at that energy level is occupied? To do this we have to also determine what is the probability that we are at a specific $\overline{N}$, given that the system can exhibit a variety of different $\overline{N}$.

The probability that the system is displaying the individual microstate $\overline{N}$, denoted by p($\overline{N}$)is given by the ratio of the number of ways in which such a microstate can be achieved, denoted by ω($\overline{N}$), to the total number of ways in which all possible microstates of the system can be achieved, denoted by ω_{Total},

i.e. $$p(\overline{N}) = \frac{\omega(\overline{N})}{\omega_{Total}}$$

Therefore the total probability that a state at energy level ε_i is occupied, $f(\varepsilon_i)$, is given by

$$f(\varepsilon_i) = \sum_{\overline{N}} f(\varepsilon|\overline{N})\, p(\overline{N}) = \frac{1}{\omega_{Total}} \sum_{\overline{N}} f(\varepsilon|\overline{N})\omega(\overline{N})$$

The above equation takes into account all of the factors necessary to make the calculation of $f(\varepsilon_i)$ complete. This equation handles the situation in its entirety.

Our analysis is based on the idea that there is no inherent preference for any one individual arrangement over the other, i.e. they are all equally probable. However the arrangements differ in the number of ways they can be arrived at, and therefore a snapshot of the system will tend to show the arrangements that can be arrived at more often. Therefore if one arrangement can be arrived at in ten different ways, and another arrangement can be arrived at in a hundred different ways, then the second arrangement is ten times more probable than the first.

For large systems, the amount of information required, as well as the amount of computation required to fully calculate all the terms of the above expression, can be unacceptably large. Fortunately, large systems provide us with the simplification recognized in statistical mechanics, that the most probable microstate swamps the probability of all of the other microstates combined. In other words, what we will find is that one of the microstates, or a specific $\overline{N}$, occurs vastly more often than the rest. This is the most probable microstate, which we will designate as $\overline{N_0}$.

The $\omega(\overline{N}_0)$ corresponding to this most probable microstate is so large that it is larger than the sum of all of the possible ways in which all of the rest of the microstates can be arrived at, combined.

Stated mathematically,

$$\omega(\bar{N}_0) \approx \omega_{total}$$

The ω_{total} will, to be exact, contain contributions from all of the other possible microstates, but these contributions will be insignificant.

Therefore, from the equation above we find that when we identify $\bar{N}_0$,

$$f(\varepsilon) \approx f(\varepsilon|\bar{N}_0)$$

Since, as indicated earlier, $f(\varepsilon|\bar{N}_0)$ is simply $\frac{n_i}{s_i}$ when $\bar{N} = \bar{N}_0$, all other contributions can be neglected.

To better understand the different terms we have used so far in this discussion, the following numbers, for $\bar{N}$, $f(\varepsilon|\bar{N})$ and $p(\bar{N})$ have been arbitrarily selected to indicate a few different possibilities. The numbers as well as the associated calculations below can be looked at purely as an illustrative example:

$\bar{N}$	$f(\varepsilon\|\bar{N})$	$p(\bar{N})$
$\bar{N}_1$	90%	0.00010
$\bar{N}_2$	10%	0.00090
$\bar{N}_3$	40%	0.99900

In view of $p(\bar{N})$ of $\bar{N}_3$ being 0.99900, it is designated as the most probable microstate, $\bar{N}_0$

Therefore:

$$f(\varepsilon) = \sum_{\bar{N}} f(\varepsilon|\bar{N}) \times p(\bar{N}) \approx f(\varepsilon|\bar{N}_0) \approx 40\%$$

Even though for $\bar{N}_1$, the $f(\varepsilon|\bar{N})$ is 90%, it has virtually no impact on the final $f(\varepsilon)$ since the $p(\bar{N})$ corresponding to it is negligible. The final value of $f(\varepsilon)$ is dominated by the contribution of the most probable microstate, $\bar{N}_3$, which, as indicated above, is therefore also designated as $\bar{N}_0$.

Therefore, even though we are faced with a computationally unwieldy expression, we can conveniently choose to ignore most of the computation involved, and focus simply on the computation that maximizes $\omega(\bar{N})$. In the process of computing the maximum possible value for $\omega(\bar{N})$, we will be able to identify an expression for $\frac{n_i}{s_i}$, which corresponds to the maximum possible value of $\omega(\bar{N})$. This expression for $\frac{n_i}{s_i}$, for all practical purposes, is treated as the overall probability that a state at energy level ε_i is occupied. It is important

to note that we have neglected a vast majority of the terms from our original expression for $f(\varepsilon_i)$. It is just that we are accepting that all of those terms, taken together, have negligible impact when compared to the impact of the most probable state, therefore we focus entirely on the probability of occupancy of an energy level when the system is in its most probable state and have stated that this is the probability of occupancy of the energy level even when all of the other states are fully taken into account.

If we denote the most probable microstate of the system as $\bar{N}_0$, then mathematically the approximation we are making can be stated as:

$$f(\varepsilon) \approx f(\varepsilon|\bar{N}_0)$$

Our task is therefore to determine the $\bar{N}_0$ which results in the maximum possible value for the corresponding $\omega(\bar{N}_0)$.

The method for identifying $\omega(\bar{N}_0)$ is to write a general expression for $\omega(\bar{N})$ and to look for the conditions that maximize $\omega(\bar{N})$. A given $\bar{N}$ consists of the following:

Energy levels:	ε_1,	ε_2,	ε_3,	ε_4,...	ε_i
Number of states:	s_1,	s_2,	s_3,	s_4,…	s_i
Number of particles:	n_1,	n_2,	n_3,	n_4,…	n_i

The number of ways to accomplish a given $\bar{N}$, denoted by $\omega(\bar{N})$ is given by the product of the number of ways in which n_1 particles can be arranged in $s_1$states times the number of ways in which n_2 particles can be arranged in $s_2$states…times the number of ways in which n_i particles can be arranged in s_i states.

The number of ways in which n_i particles can be arranged in s_i states is denoted by $\Omega(n_i)$

Stated a little bit more elaborately, at a given energy level ε_i, the number of ways in which we can place n_i identical and indistinguishable particles in the available s_i states, with no more than one particle per state (Pauli's exclusion principle), can be expressed mathematically as:

$$\Omega(n_i) = \frac{s_i!}{n_i!(s_i - n_i)!}$$

The above expression can be thought of as the number of combinations possible with n_i identical and indistinguishable particles, and (s_i-n_i) identical and indistinguishable empty states, in a total of s_i states.

For a specific $\overline{N}$, therefore:

$$\omega(\overline{N}) = \Omega(n_1) \times \Omega(n_2) \times \Omega(n_3) \ldots = \prod_i \Omega(n_i) = \prod_i [\frac{s_i!}{n_i!(s_i - n_i)!}]$$

Our intent is to maximize $\omega(\overline{N})$ subject to the two constraints placed on the system which are that the total energy of the system is a constant, and that the total number of particles in the system is a constant.

i.e.

$$\sum \varepsilon_i n_i = constant$$

and

$$\sum n_i = constant$$

In this process of maximizing $\omega(\bar{N})$, subject to the above two constrains, we will find $\frac{n_i}{s_i}$ in the most probable $\bar{N}_0$, which is the $f(\varepsilon|\bar{N}_0)$ and hence the $f(\varepsilon)$ that we seek, since contributions from all other terms will be negligible.